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Building a boundary-spanning service for coopetition

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ABSTRACT

Coopetition is a revolutionary mindset, combining cooperation and competition and allowing companies to work with potential collaborators based on their preferences. It challenges the concept of a supply chain, where companies occupy different positions within a layered structure. Relationships in a supply chain set boundaries for the different collaborators. In contrast, the partners in coopetition can be either collaborators or competitors, and the relationship can evolve over time. To support coopetition, a platform is required for dynamic relationships to develop and for various types of collaboration to occur. This platform enables companies to effectively form networks via a boundary-spanning service, and thus gain competitive advantages. A boundary-spanning service is an intelligent intermediary, providing information, contingencies, and role routines with discretionary powers under constraints. This study demonstrates a process of collaborative purchasing whereby companies negotiate over different offers to increase their utilities using boundary-spanning agents.

1. Introduction

Collaborative networks allow businesses to extend collaboration agreements beyond the traditional layered supply chain to include various situations and participants. Networks allow for collaborations between members of a supply chain who are directly connected, on the same layer of the chain, or who are further upstream or downstream. Traditionally, forming tighter supply chain relationships results incompetitive advantages by enabling the quick responses that are necessary to achieve strategic goals (Daft & Weick, 1984), such as lower inventory levels. Companies on the same layer of a supply chain are normally considered as competitors. The relationships therefore form boundaries to the layered network.

However, when a collaborative network is established, the competitive relationship becomes blurred. A company must maintain good relationships not only with suppliers and buyers, but also with their same-layer competitors. This revolutionary approach is termed coopetition, and is a combination of cooperation and competition (Brandenburger & Nalebuff, 1996). The coopetition relationship can create different types of cooperation, which ultimately benefit a company; for example, a company can work with its same-layer competitors to obtain a better volume discount from a seller.

The establishment of a collaborative network allows the possible modes of cooperation and competition to be extremely dynamic. Suppliers, buyers, and other collaborators can be easily located, maximizing efficiency. In this study, we examine the use of purchasing data in a cloud-based collaborative network. Boundary-spanning agents allow participants to identify other collaborators and negotiate with them. These boundary-spanning agents—also known as boundary spanners or boundary-spanning individuals—are intelligent intermediaries. Their functions are to (1) process and filter information; (2) provide for contingencies; (3) create the role routine; (4) possess some degree of discretionary power; (5) take on different technological and boundary roles (mediating, an inter-dependent role, or intensive interaction) for different types of organizations; and (6) deal with environmental constraints (Aldrich & Herker, 1977; Ancona & Caldwell, 1992; Tushman & Scanlan, 1981).

In our study we have designed an intelligent, boundary-spanning agent that provides decision-making support and can automatically search for, negotiate, and evaluate solutions for a participant (Barbati, Bruno, & Genovese, 2012). A collaboration support tool is not a type of collaborative software or "groupware" (Ellis, Gibbs, & Rein, 1991) enabling different individuals involved in a common task to achieve their goals. It instead works at the firm level, similar to a negotiation support system (NSS), such as active collaboration and negotiation frameworks (ACNF), where active documents with embedded business rules are used to adapt to different collaborative strategies in a business-to-business (B2B) environment (Du & Chen, 2007). It is also comparable to DSS agents used to reason cooperatively with other agents in a process of interactive constraint satisfaction (Wang, Liao, & Liao, 2002). A collaboration support tool should be more than an NSS, in which the primary concerns are the negotiations with other

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participants to obtain a lower price or achieve other goals. An effective tool should create a collaborative working environment based on a collaborative network, but also search for, evaluate, and recommend potential partners based on the collaborative strategies of firms.

2. Collaborative networks

A traditional supply chain integrates the entire production line, from upstream component suppliers to downstream consumers. The relationships between the upstream and downstream parties are linear, regardless of the numbers in each layer. Supply chain collaboration is currently an indispensable strategic business tool (Chopra & Meindl, 2003; Ramanathan, 2014). Extension of the supply chain in turn extends links vertically and/or horizontally. These types of extensions take advantage of Web technology to form a network structure known as a collaborative network (CN). This can be established in many different formats, and can be either mobile or stationary. It can integrate information, products, and services. Examples of CN applications include vendor management inventories (VMI), information sharing between P&G and Wal-Mart, retailer–supplier collaboration at Target, transportation and inventory enhancement at Unilever, and design cycle reduction at Adaptec (Turban, Lee, King, & Liang, 2010). ACN integrates various business functions including demand planning, planning and scheduling, order management, product development, vendor management, and sales support. The quality of a collaboration can be measured through various improvements: (1) better relationship management with unidirectional partners; (2) improved integration of business functions; (3) a stronger foundation of trust in knowledge and information sharing; and (4) an atmosphere more conducive to establishing a collaborative culture (Li & Du, 2005).

Collaboration is markedly more effective when a network is cloud-based where an innovative platform providing highly scalable Web-based services (Buyya, Yeo, & Venugopal, 2008; Foster, Zhao, Raicu, & Lu, 2008; Jula, Sundararajan, & Othman, 2014). The geography of computation shifts from local computers (Hayes, 2008) to the broader environment of the cloud, where the on-demand services can be optimized by participants (Armbrust et al., 2009). Based on the architecture, services can be divided into software, platforms, or infrastructure (Grossman, 2009; Rhoton, 2010). The software can comprise a wide range of applications such as CRM, ERP, or email, which can be purchased from service providers, and also shared. Various commercial platforms such as Google, Amazon, and Apple apps can be used, as can tools such as specific programming languages or development environments. Users can determine the level of investment (monetary and technological) required before deciding whether to purchase services. Downstream infrastructure services can extend beyond the adoption of hardware to areas such as power, real estate, computation, memory, storage, or virtualization (asking other users to act on one's behalf). There are many possible adoption modes available, and toolkits can be used to evaluate these using simulated solutions (Calheiros, 2011) or cost models (Khajeh-Hosseini, Greenwood, Smith, & Sommerville, 2012).

3. Boundary-spanning service

3.1. Framework of boundary-spanning service

A collaborative network can involve numerous participants, such as production firms, service providers, non-profit organizations, and governmental units. A boundary-spanning service is responsible for sharing data between all participants, and in the cloud is a collaborative support tool designed to help organizations gather, store, and access information and knowledge easily, and also to provide them with decision-making support within collaborative networks. The functions of the boundary-spanning service agent include those of profile manager, relationship manager, and service engine (shown in

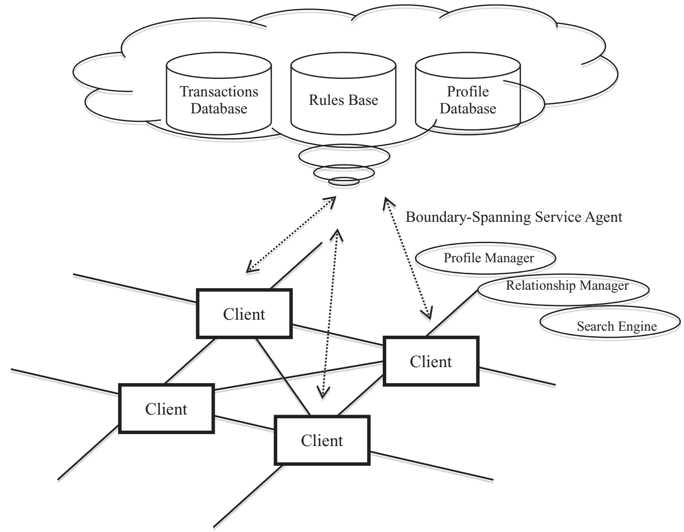

Fig. 1. The boundary-spanning service.

Fig. 1). The profile manager provides basic information about the parties involved and the corresponding preference vectors describing their inclination toward other parties. The corresponding preference vectors can be updated on the basis of past transactions, such as previous purchasing information. The preference P is represented as: $P = (p_1, p_2, \ldots, p_m)$, where $\sum_{i=1}^{m} p_i = 1$ and i is an involved party.

The main role of the relationship manager is to compute, store, and update relationships, i.e. the preference vector established by the profile management function. It is dynamic in nature depending on the contracts, negotiation, and other factors between two companies in coopetition of the collaborative network. The service engine is the core feature of the boundary-spanning service. It can enable a company to search for and contact service providers, and negotiate and recommend collaborators according to different criteria, such as price and delivery date. Recommendations are not limited to potential cooperators (e.g., sellers to a purchaser), but can also encompass competitors (e.g., those looking to buy the same items) if cooperation can enhance utility (e.g., giving a better discount for a cooperative quantity purchase).

3.2. Boundary-spanning service design

In a cloud-based collaborative network, clients (buyers or sellers) collaborate with others by delivering active documents through the boundary-spanning service agents. These active documents, written in XML, are embedded with business logics that can adapt the content to different collaborative strategies. Active documents, as opposed to passive documents that contain only content and a structured presentation, include dynamic and interactive components that are similar to the application interfaces for human use (Nam, Jang, & Bae, 2003). For example, an active document can be used by a buyer to search, evaluate, and recommend potential partners based on the collaborative strategies. This design means that the document's function is no longer limited to serving as a medium for data storage. In fact, if advanced inferences are built-in, the document can also process applications.

Following the predefined instructions and threshold values, which are written in the active document, a boundary-spanning service agent can inquire, collaborate, and negotiate with the boundary-spanning service agents of other sellers or buyers in a cloud-based collaborative network. For example, when a participant asks for a buying service, the following information is provided: the purchasing target is 100 laptops; the product specifications include the size

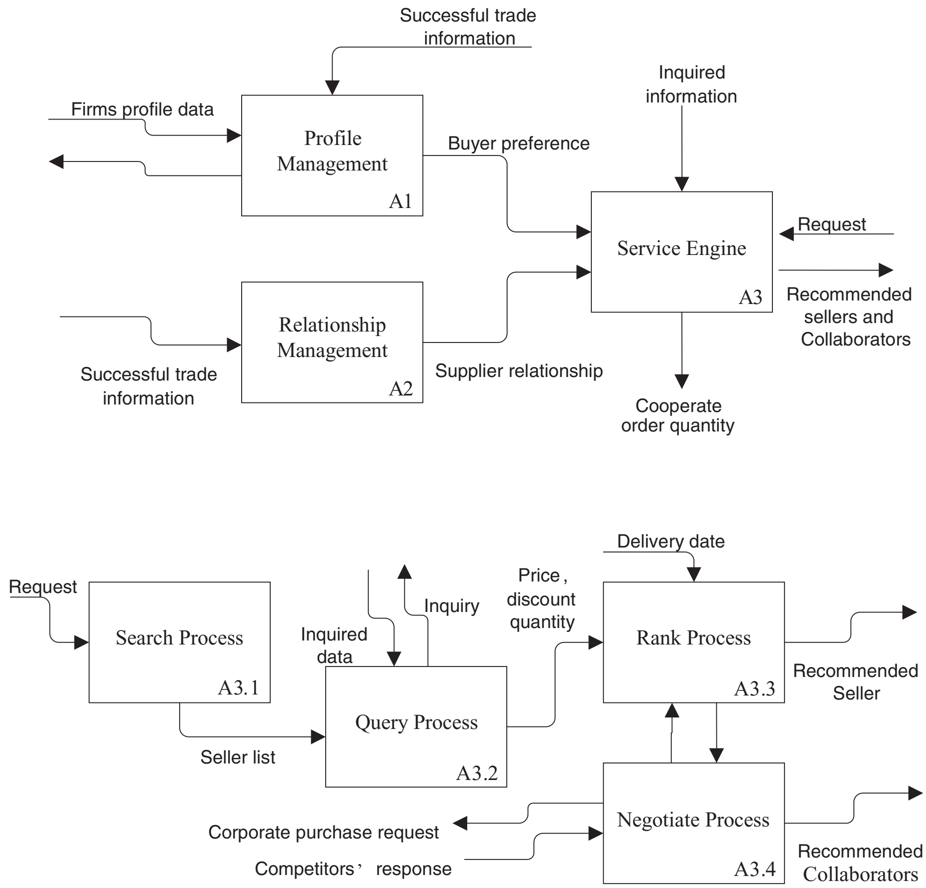

Fig. 2. The boundary-spanning service process in IDEF0 representation.

of the hard drive and memory; the requirements are the delivery date and others; the constraints are acceptable price range and others; and the business logic and the CPU speed should be decided before determining the motherboard choice. This information is kept in an active document and sent to other clients through the collaborative network using a boundary-spanning service agent.

Specifically, the boundary-spanning service has three parts, illustrated in Fig. 2: a relationship management function (A1), a profile management function (A2), and a service engine (A3). The service engine is the core service and interacts with the other two functions. The service engine process can also be divided into four subprocesses: search (A3.1), query (A3.2), ranking (A3.3) and negotiation processes (A3.4).

(1) The *search process* is the process through which a request is received from a company, and the boundary-spanning service then searches qualified collaborators (buyers or sellers, at this stage). We will later use a purchase request sent by a buyer to illustrate the procedure.
(2) The *query process* issues a request for detailed information such as prices and discount quantities to potential collaborators from the list generated by the search process (A3.1). When a purchase request is received by the boundary-spanning service, the service engine sends inquiries to all sellers in the network, together with the criteria indicated in the request, such as price and quantity (step A3.1). A seller can then respond to the inquiry and provide a quantity discount as an incentive to the buyer to make a larger quantity purchase. The quantity discount may also come with certain constraints, such as a purchase period or delivery batch. Buyers who can afford to make a bulk purchase or to wait for a certain period will benefit from the better price. Negotiations can start between buyers and sellers to obtain a better price, between buyers and buyers to purchase a larger aggregate volume, or between sellers and sellers to absorb a larger volume of purchase orders.
(3) The *ranking process* computes and ranks the relative utilities if the company decides to work with the identified collaborators.

In our case, a seller has a large capacity and offers a quantity discount. We assume that the discount is as follows (Gurnani, 2001):

$$c = \begin{cases} c_0 & 0 \le q \le q_1 \\ c_1 & q_1 \le q \le q_2 \\ \ldots & \ldots \\ c_{min} & q \ge q_{max} \end{cases}, \qquad (1)$$

where the initial unit price c_0 is the price announced by the supplier if the order quantity is less than q_1; the unit price is c_1 when the order quantity is between q_1 and q_2; and c_{min} is the minimum price the supplier can offer when the order quantity is larger than q_{max}.

In this paper, we consider coopetition decision making as a multi-criteria decision-making (MCDM)—or multi-attribute decision-making—problem in which numerous criteria are considered as attributes for decision making. In the real world, choosing a seller or a cooperator involves a number of factors. The service engine uses a fuzzy MCDM method and company profiles to evaluate the utilities if the buyer works with the potential sellers (Deng & Chan, 2011; Yang, Chiu, Zeng, & Yeh, 2008). The matching between buyers and sellers involves a number of factors. Each buyer has his/her own priority for criteria selection. This is an MCDM process that considers multiple attributes for ranking sellers, such as unit price, delivery date, discount, quality level, etc. To solve the problem, attributes such as quality level can be represented by fuzzy functions because they do not have clear quantitative boundaries. In this way, we can allow descriptive attributes—such as bad, medium, or good—for quality level. Then, membership functions can be used to represent the degree of truth as an extension of valuation. Thus, fuzzy MCDM methods can be used for matching purposes.

If a decision space S contains n sellers who are registered in the cloud and selling the product then $S = \{s_1,s_2,\ldots,s_n\}$. Let $f_i(s_j)$ denote the value of attribute $f_i(s)$ of seller s_j and $f_i(s) = (f_1(s_j), f_2(s_j), \ldots, f_m(s_j))^T$ represent all attribute values of seller s_j. Here, the attributes of the sellers include not only operational attributes $(f_1(s_j), f_2(s_j), \ldots, f_{m-1}(s_j))^T$ such as price, discount, delivery date, quantity, and yield rate, but also the records of previous successful trades between the buyer and sellers over a certain period, $f_m(s_j)$. That is, when a buyer is satisfied with a trade, their willingness to cooperate in the future is recorded in$f_m(s_j)$indicating a successful trade experience (Jackson, 1985; Selnes, 1998). If the membership function of $f_i(s)$ is represented as $u_{\tilde{f}_i}(s)$, the MCDM problem can be described as $\max_{s_j\in S}\{u_{\tilde{f}_i}(s_j)\}$, where $u_{\tilde{f}_i}(s_j) = (u_{\tilde{f}_1}(s_j), u_{\tilde{f}_2}(s_j), \ldots, u_{\tilde{f}_m}(s_j))^T \in [0,1]^m$.

There are three types of seller's attributes: cost, benefit, and fixed attributes. Cost attributes such as price and delivery data have a negative relationship with the buyer's preferences, whereas benefit attributes such as the yield rate are positively related to preferences. Fixed attributes such as quantity have a non-monotonic relationship with preferences. Buyers show certain preference values, known as objective values, in relation to these attributes. The degree of membership of all attributes can be calculated by formulae (2), (3), and (4).

$$u_{\tilde{f}_i}(s) = \frac{\sup\left\{f_i(s)\right\} - f_i(s)}{\sup\left\{f_i(s)\right\} - \inf\left\{f_i(s)\right\}}, \tag{2}$$

$$u_{\tilde{f}_i}(s) = \frac{f_i(s) - \inf\left\{f_i(s)\right\}}{\sup\left\{f_i(s)\right\} - \inf\left\{f_i(s)\right\}}, \tag{3}$$

$$u_{\tilde{f}_i}(s) = \frac{f_i^*(s)}{f_i^*(s) + \left|f_i(s) - f_i^*(s)\right|}, \tag{4}$$

where $\inf\{f_i(s)\}$ is the lower boundary of $f_i(s)$ in S and $\sup\{f_i(s)\}$ is the upper boundary of $f_i(s)$ in S.

We assume that the number of successful trades between two firms can be used to indicate how close their relationship is. Therefore, $u_{\tilde{f}_m}(s)$ is the relationship value calculated by formula (5):

$$u_{\tilde{f}_m}(s) = \frac{f_m(s)}{\sup\{f_m(s)\}}, \tag{5}$$

where $\sup\{f_i(s)\}$ is the upper boundary of $f_i(s)$ and is the largest number of successful trades between the firm and one of its partners.

Based on the membership values of all criteria, the relative utility of the firm, if it chooses seller j, can be calculated using formula (6) and the buyer's preference vector. By comparing $U(s_j)$, all of the sellers can be ranked.

$$U(s_j) = \sum_{i=1}^{m} p_i \mu_{\tilde{f}_i(s_j)}. \tag{6}$$

(4) The *negotiation process* involves a party or multiple parties engaging in negotiations. The service agent negotiates with the potential collaborators and explores other possibilities (for example, working with competitors, such as buyers, in a collaborative purchase, as mentioned) based on the relative utilities computed during the ranking process (A3.3).

All buyers in the cloud that might purchase the same product during the same period use the same boundary-spanning service, so their preferences and utility information are all stored in the database. The boundary-spanning agent can therefore identify the buyers that may be interested in buying the same product from the same seller. Then, the *negotiation process* sends cooperative purchase requests to those buyers. In this case, the original buyer is no longer a sole buyer, but faces a new competition–cooperation decision that includes other buyers expressing interest in buying goods from the same seller. In general, if the boundary-spanning service includes X buyers and Y sellers who may be involved in business transactions for the same product, we use U_{x-y} to represent the relative utility of buyer x if it purchases a product from seller y. There are therefore four possible relative utilities involved if buyer B_x wants to purchase a product from seller S_y: U_{x-y} (the relative utility if B_x purchases the product from S_y at the initial price c_0); U'_{x-y} (the relative utility if B_x increases its order quantity to q_0 to obtain the quantity discount r from S_y); U''_{x-y} (the relative utility if B_x cooperates with other buyers to obtain the quantity discount r from S_y). A participant can evaluate which proposal is the most acceptable.

To simplify the scenario, we use an aggregate purchase for illustration. Here, a buyer has three possible choices when buying a product through the boundary-spanning service. He/she can make a decision according to his/her relative utilities: (1) to buy the product at the initial price c_0; (2) to increase the order quantity to q_0 to obtain the quantity discount r; or (3) to cooperate with competitors (other buyers) to obtain the quantity discount r. The utility function of buyer B_x is therefore

$$\max\begin{Bmatrix} buy & directly \\ increase & quantity \\ cooperation & purchase \end{Bmatrix} = \max\begin{Bmatrix} Max\{U_{x-1}, U_{x-2}, \ldots, U_{x-Y}\} \\ Max\{U'_{x-1}, U'_{x-2}, \ldots, U'_{x-Y}\} \\ Max\{U''_{x-1}, U''_{x-2}, \ldots, U''_{x-Y}\} \end{Bmatrix} \tag{7}$$

A buyer's relative utility U'_{x-y} should generally be smaller than U''_{x-y}, as cooperative purchasing can reduce the unit price without changing the order quantity to q_0. The problem is whether a buyer can find other potential buyers to work with to increase its utility to U''_{x-y}. To illustrate, suppose that B_{x1} is willing to buy a product in the boundary-spanning service. We let $U_{\max} = Max\{U_{x1-y}, U'_{x1-y}\}$ denotes for the maximum utility without cooperative purchase. Buyer B_{x1} will be interested in collaborating with other buyers to cooperative purchase from sellers with $U''_{x1-y} > U_{Max}$.

Considering that B_{x1} and B_{x2} can buy the product from other sellers, it is only if

$$\max\begin{Bmatrix} Max\{U_{x1-1}, U_{x1-2}, \ldots, U_{x1-Y}\} \\ Max\{U'_{x1-1}, U'_{x1-2}, \ldots, U'_{x1-Y}\} \\ Max\{U''_{x1-1}, U''_{x1-2}, \ldots, U''_{x1-Y}\} \end{Bmatrix} = U''_{x1-y} \quad \text{and}$$

$$\max\begin{Bmatrix} Max\{U_{x2-1}, U_{x2-2}, \ldots, U_{x2-Y}\} \\ Max\{U'_{x2-1}, U'_{x2-2}, \ldots, U'_{x2-Y}\} \\ Max\{U''_{x2-1}, U''_{x2-2}, \ldots, U''_{x2-Y}\} \end{Bmatrix} = U''_{x2-y}$$

that B_{x1} and B_{x2} will purchase from S_y together. Thus, U''_{x1-y} and U''_{x2-y} depend on B_{x1} and B_{x2} reaching a compromise. If the sum of these buyers' order quantities is greater than q_0, both can obtain their primary order quantities. However, if the sum of their order quantities is still smaller than q_0, they must increase their combined order quantity to q_0 to obtain the discount. In this situation, the sharing of the shortfall depends on the results of bargaining between the two buyers (or more than two buyers in a real-world situation). Buyers compromise with others throughout the bargaining process according to their utility functions.

Note that the relative utility functions are used to model the negotiators' utilities, and the initial solution is obtained by intersecting the utility functions of both parties. The negotiation process allows both parties to arrive at the first settlement quickly and efficiently if

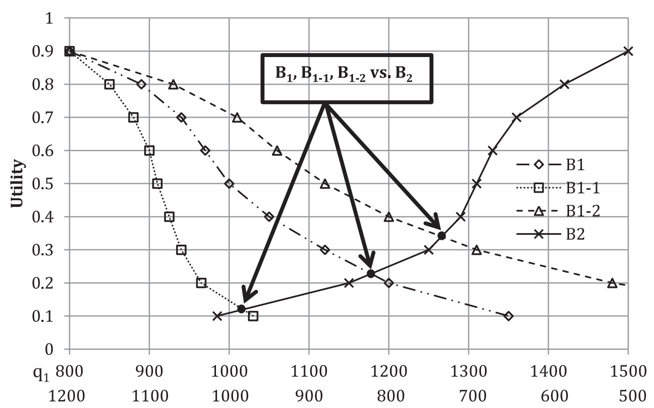

Fig. 3. Differences in settlement when utility functions are changed.

there is a positive bargaining zone. Unlike negotiation between buyers and sellers, which has been fully discussed in previous automatic negotiation systems, our study focuses on negotiation between buyers in coopetitive situations in a collaborative network. It is assumed that sellers in the collaborative network offer their relatively stable quantity discount strategies.

Once the buyers' utility functions have been constructed, the consensus value of each criterion (mainly quantity) between the buyers can be located by finding the intersection of their utility functions. If the intersection points can be found for all of the utility functions of the buyer and other buyers, then the value of the multi-criteria relative utility function U''_{x-y} for the buyer can be calculated. If no intersection can be located for a criterion, then the expectations of the current buyers are mutually exclusive and the parties will not be able to reach a cooperative purchase agreement. In this case, if the number of times that a buyer has proposed to other buyer has reached the user defined maximum, then the buyer will be asked to compromise and to reconstruct its utility functions so that intersection points can be found.

As an example, take two buyers (B_1, B_2) that wish to purchase LCD monitors in a collaborative network. The order quantity (q_1) of buyer B_1 is 800, and the order quantity (q_2) of buyer B_2 is 500. The seller (S_1) offers a discount when the quantity is greater than 2000. The shortfall Δp is 700. Fig. 3 shows the settlements that are derived from different utility functions that a buyer changes when it negotiates with the other buyer. When buyer B_1 changes its preference from the utility functions of buyer B_1 to those of buyer B_{1-1} (more sensitive to order quantity) and buyer B_{1-2} (less sensitive to order quantity), the settlement point also changes. This also explains why, when the buyer's negotiation agent cannot reach an agreement about the utility threshold with other buyer's agent, changing the utility functions can help to reach a conclusion.

The negotiation engine therefore helps to locate potential sellers for a buyer and find possible buyers (who can also be competitors) to cooperate with in purchasing deals, and also suggests actions that can possibly improve the utilities of both buyers and sellers.

Once a purchase is confirmed, the trade record is used to update both the preference database and the relationship database. That is, when the buyer completes a successful trade with one or more suppliers, the corresponding successful trade number $f_m(s_j)$ is increased (by one in this simplified case) to indicate his/her preference. The relationship indicator between buyers is also updated if they successfully cooperate in purchasing from the same supplier. The preference vector is changed after the buyer selects seller s_j as set out below. In short, the procedure is:

Input

Firm F_k's preference: $p^{F_k} = (p_1^{F_k}, p_2^{F_k}, \ldots, p_i^{F_k}, \ldots, p_m^{F_k}),\ 1 \le i \le m.$
The attributes of selected seller j: $f(s_j) = (f_1(s_j), f_2(s_j), \ldots, f_m(s_j))^T,\ 1 < i < m$
Output
Updated preference: $\tilde{p}^{F_k}$.
Updating preference
Let ε be the update parameter indicating the sensitivity of the user preference updating.
Compute $\tilde{p}_i^{F_k} = p_i^{F_k} + \varepsilon[\mu_{\tilde{f}_i}(s) - p_i^{F_k}]$ (for i = 1 to m)
The updated preference: $\tilde{P}^{F_k} = (\tilde{p}_1^{F_k}, \tilde{p}_2^{F_k}, \ldots, \tilde{p}_i^{F_k}, \ldots, \tilde{p}_m^{F_k}),\ 1 < i < m.$

4. System demonstration

4.1. Initial setting

This section demonstrates the use of the boundary-spanning service for a group purchase. Here, we assume that all sellers have a simple discount quantity strategy (Fazel, Fischer, & Gilbert, 1998):

$$c = \begin{cases} c_0 & 0 < q < q_0 \\ rc_0 & q_0 \le q \end{cases},$$

where q_0 denotes the discount quantity and r is the discount rate. We randomly generate preference vectors and order quantities to represent five buyers for a particular product. To obtain the buyers' preferences, we use a pair-wise comparison by first generating weightings reflecting the relative importance of criterion f_1 against other criteria, using a discrete uniform distribution (1/5, 1/4, 1/3, 1/2, 1, 2, 3, 4, 5). A weight (w_{ij}) greater than 1 indicates that criterion p_i is more important than criterion p_j. That is, $w_{ij} = 2$ means criterion p_i is slightly more important than criterion p_j, whereas $w_{ij} = 5$ means criterion p_i is dominant over factor p_j. Similarly, if w_{ij} is less than one, criterion p_i is less important than p_j. Based on the transitivity of pair-wise comparisons, the relative importance of each criterion is calculated to form a matrix. A preference vector can then be calculated to represent the buyers' relative preferences among the criteria. Buyers' order quantities are randomly generated based on a uniform distribution over the interval [500, 1000]. The generated preference vectors and order quantities for the five buyers are listed in Table 1.

For demonstration purposes, we select sellers using five criteria: price (f_1), delivery date (f_2), yield rate (f_3), quantity (f_4), and the relationship between buyer and seller (f_5). Price (f_1) is uniformly distributed over the interval [100, 120]. Ten sellers are randomly generated. A discrete uniform distribution is used to randomly generate a distance matrix for the delivery dates, ranging from one to five days. The yield rate is generated from the uniform distribution over the interval [0.6, 1.0]. In reality, every seller would have a different strategy for discount quantities (q_0) and discount rates (r), but to simplify the demonstration we assume that all sellers have the same quantity discount strategy with $q_0 = 1500$ and $r = 0.9$. The quantity $(q_0 = 1500)$

Table 1
Five generated buyers with preferences based on five criteria and their initial order quantities.

Buyer ID	p_1	p_2	p_3	p_4	p_5	Quantity
B1	0.092735	0.273341	0.299404	0.146159	0.18836	500
B2	0.262384	0.049402	0.156304	0.360242	0.171669	700
B3	0.170618	0.349426	0.018148	0.392761	0.069046	600
B4	0.280167	0.239569	0.285943	0.108257	0.086064	800
B5	0.336686	0.153567	0.073862	0.19722	0.238665	600

Table 2
Criteria values of buyer *B1* against those of 10 generated sellers.

Seller ID	f_1	f_2	f_3	f_4	f_5
S1	108	4.00	0.76	1500	0
S2	118	5.00	0.80	1500	0
S3	114	2.00	0.95	1500	0
S4	105	3.00	0.64	1500	0
S5	104	4.00	0.86	1500	0
S6	110	2.00	0.77	1500	0
S7	115	1.00	0.91	1500	0
S8	101	5.00	0.83	1500	0
S9	105	2.00	0.74	1500	0
S10	109	5.00	0.86	1500	0

Table 3
Buyer B1's utility when it purchases from each seller.

Seller ID	Buyer utility		
	U_{1-y}	U'_{1-y}	U''_{1-y}
S1	0.270	0.224	0.303
S2	0.216	0.163	0.257
S3	0.482	0.430	$\underline{0.527}$
S4	0.254	0.201	0.269
S5	0.361	0.309	0.373
S6	0.373	0.321	0.408
S7	$\underline{0.499}$	0.447	$\underline{0.545}$
S8	0.303	0.251	0.303
S9	0.372	0.319	0.386
S10	0.292	0.240	0.324

is chosen to allow the possibility of cooperation among buyers when their aggregate initial order quantity exceeds the amount required to obtain a seller's discount. We also initially assume there is no trading history between buyer B1 and all sellers. The initial value of relationship (f_5) is therefore set at 0. The 10 sellers generated are shown in Table 2.

The demonstration process can be described as follows:

Step 1: Buyers B = {Bi, i = 1–5} search for and inquire from all sellers S = {Sj, j = 1–10} in the collaborative network.
Step 2: Buyers calculate their U_{i-j}, U'_{i-j} and U''_{i-j}, respectively.
Step 3: Buyers get their U_{max} and find the sellers they are interested in purchasing from other buyers.
Step 4: Buyers send corporate purchase requests to other buyers who are interested in the same seller.
Step 5: Buyers negotiate with other buyers if the sum of buyers' order quantities is smaller than the seller's discount quantity q0.
Step 6: Buyers recalculate their U_{i-j}, U'_{i-j} and U''_{i-j} and choose the best seller and cooperator.

4.2. Negotiation and cooperative purchase

Suppose that one buyer makes only one purchase during a certain period (purchase period t). The utilities for buyer B1 at $t = 1$ are shown in Table 3. Here, we take U_{1-1} of B1 for the purpose of further illustration. The attributes of S1 are ($f_1(s_1)$, $f_2(s_1)$, $f_3(s_1)$, $f_4(s_1)$, $f_5(s_1)$) = (108, 4, 0.76, 1500, 0). Based on formulae (2), (3), (4), and (5), the membership degrees of all attributes can first be calculated as $(\mu_{\tilde{f}_1}(s_1), \mu_{\tilde{f}_2}(s_1), \mu_{\tilde{f}_3}(s_1), \mu_{\tilde{f}_4}(s_1), \mu_{\tilde{f}_5}(s_1)) = (0.50, 0.25, 0.39, 1,0)$. We can then use formula (6) to obtain $U_{1-1} = 0.275$ with the buyer's preference vector.

Table 3 shows that without the quantity discount, the best seller for B1 is S7 ($U_{1-7} = 0.449$). There are two sellers (S7, S3) with whom B1 may be interested in working with, in collaboration with more buyers, as the corresponding utilities U''_{1-7} and U''_{1-3} are greater than U_{1-7}. Thus, the agent of Buyer B1 will send corporate purchase requests to other buyers who are also interested in the same sellers (S7 or S3). After all of the buyers' agents send their requests, Buyer B1 receives two requests, one each from Buyers B3 and B4. The agent then finds an initial settlement based on the utility functions of all buyers, as shown in Table 4. Buyer B4 is interested in S3, whereas Buyer B3 wants to get a discount from S7. Both of the buyers cannot satisfy Buyer B1 (Total utility <0.499) after iteration 1 and are asked to resubmit their proposals in iteration 2. The scale of the negotiation zone is set as 5% of the bargaining zone. After iteration 3, Buyers B3 and B4 are recommended to Buyer B1 based on their total utility values.

In the collaborative network, efficient information acquisition makes it possible for each company to test different proposals for improving its utility. We assume that buyers are rational and will choose the seller that brings them the highest utility; a buyer will work with other buyers to negotiate with seller(s) to enhance utility. The results for the five buyers listed in Table 1 are shown in Table 5. Buyer B1 changes to seller S3 after cooperating with buyer B4, and seller S7 eventually sells the product to B2, instead of B1.

To test for sensitivity, let $\varepsilon = 0.1$. The new buyer preferences are reported in Table 6. In this experiment, we again randomly generate order quantities for the five buyers and place purchase orders. We also let the unit prices of all sellers randomly change within a 10% range to simulate price fluctuations in the market. We repeat the test 10 times ($t = 10$) and the outcome for buyer B1 is shown in Table 7. The buyer works with B4 to buy products from the seller S3 twice, with B3 to buy from S5, and with B5 to buy from S7.

Table 4
Settlements reached by buyer B1 and other buyers using utility function.

Buyer ID	Seller ID	Iteration	Discount quantity	Proposal	Utility	Total utility
B4	S3	1	1500	780	0.482	0.435
B3	S7	1	1500	759	0.409	0.454
B4	S3	2	1500	823	0.438	0.478
B3	S7	2	1500	782	0.387	0.492
B4	S3	3	1500	846	0.427	0.527

Table 5
Results for five buyers after initial sourcing.

Buyer ID	Seller ID	Unit price	Delivery date	Yield rate	Quantity	Utility
B1	S3	102.6	2	0.95	564	0.527
B2	S7	115	3	0.91	700	0.453
B3	S5	104	5	0.86	600	0.489
B4	S3	102.6	4	0.95	846	0.494
B5	S8	101	3	0.83	600	0.506

Table 6
Updated preference vector for all buyers after a successful transaction.

Seller ID	p_1	p_2	p_3	p_4	p_5
B1	0.10284	0.276481	0.28453	0.14782	0.188329
B2	0.273495	0.043971	0.148707	0.368946	0.164881
B3	0.161889	0.336009	0.042962	0.382118	0.077023
B4	0.275009	0.242461	0.286772	0.117782	0.077977
B5	0.336686	0.153567	0.073862	0.19722	0.238665

Table 7
Results for buyer B1 after 10 rounds of sourcing.

Seller ID	Unit price	Quantity	Utility	Cooperator
S3	102.6	564	0.525	B4
S6	101.4	900	0.553	-
S7	104.3	700	0.519	-
S8	116.4	600	0.486	-
S7	102.1	800	0.541	B5
S5	102.6	600	0.525	-
S10	100.2	900	0.542	-
S3	107.3	900	0.511	B4
S7	105.8	600	0.596	-
S5	103.3	741	0.517	B3

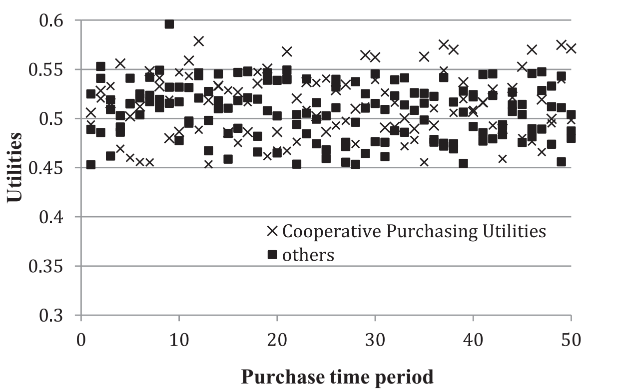

Fig. 4. Buyers' utilities when they cooperate to purchase, versus those when they do not.

The cloud provides an efficient platform for companies to conduct business. In this study, we show that a boundary-spanning service is a collaborative network that gives members a platform for adopting different strategies, either competitive or cooperative, when interacting with other members. For example, a buyer can decide to compete with other buyers for products (or vice versa for a seller) or work with them to obtain a better quantity discount. In Fig. 4, we show that in almost all purchase periods ($t = 1$–50), cooperative purchases among buyers can clearly enhance utilities (F value = 17.94, $P < 0.001$). This indicates that sharing information via the cloud provides opportunities to work with other companies, and these collaborations eventually increase utilities.

4.3. Relationships in the collaborative network

The relationships of buyers with sellers and with their competitors (other buyers in the example) are updated over time. Given that we define $f_m(s_j)$ as the number of previous successful trades between a company and seller j within a certain period, we can use a parameter $u_{\tilde{f}_m}(s)$ to indicate how close the relationship is and to display this relationship visually. The relationship network of B1, based on data obtained after10 rounds of sourcing, is shown in Fig. 3.

Relationships are very important for companies working in a collaborative network. The boundary-spanning service can help manage these relationships and track how they have evolved. Figs. 3 and 4 allow us to further explore how relationships evolve. We use $f(B_j)$

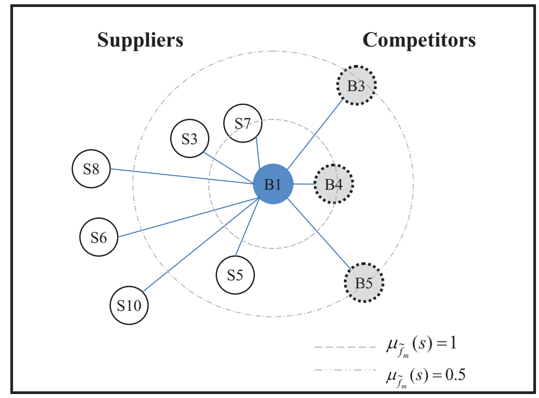

Fig. 5. Relationships of buyer *B1* with sellers and competitors (buyers) after 10 rounds of sourcing.

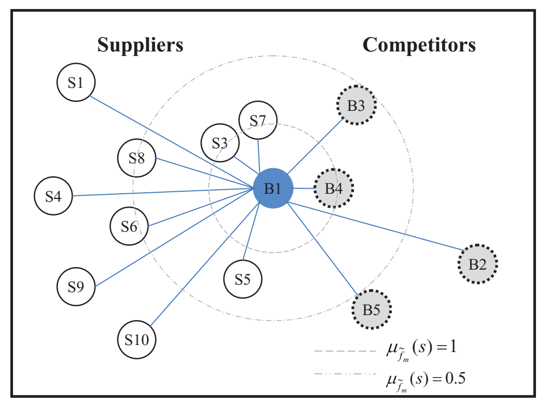

Fig. 6. Relationships of buyer B1 with others after 50 rounds of sourcing.

to represent the number of times a company and a buyer j successfully cooperate in a certain period, and $\mu(B_j)$ to denote the relationship based on formula (5). From Fig. 3, we can see that S7 ($f_5(s_7) = 3, \mu_{\tilde{f}_5}(s_7) = 1$, i.e., three trades with S7) is the seller with the closest relationship to B1, whereas B4 ($f(B_4) = 2, \mu(B_4) = 1$, i.e., works with B4 twice) is the closest competitor in the same market.

After B1 has traded for a longer period, say 50 ($t = 50$), its relationship network evolves into that shown in Fig. 5. Based on this, we can see that S7 ($f_5(s_7) = 12, \mu_{\tilde{f}_5}(s_7) = 1$) is still the closest seller to B1 and S3 ($f_5(s_3) = 11, \mu_{\tilde{f}_5}(s_3) = 0.91$) is the next closest. Sellers S1, S4, and S9 do business with B1 from time to time. Similarly, B4 has the closest relationship with B1, and B2 occasionally cooperates with B1.

Figs. 5 and 6 show that the evolution of relationships provides important information for a decision maker in a collaborative network. For a better understanding, Fig. 7 demonstrates a complete network, made up of five buyers and ten sellers. This information can be useful when dynamically managing relationships with different parties.

4.4. Quality of negotiations between buyers

In this study, we use joint utility and equality to evaluate the quality of negotiations (Du & Chen, 2007; Mumpower, 1991). A thousand settlements that occur as the result of negotiations are randomly sampled to determine the quality of the negotiations. We only run the negotiation process and simply assume that all buyers' U_{max} are 0.5. Ten samples of successful settlements that are recommended by the boundary-spanning service agents are listed. As shown in Fig. 8, the points of settlements that are recommended by the boundary-spanning service are located very close to the edge of the bargaining

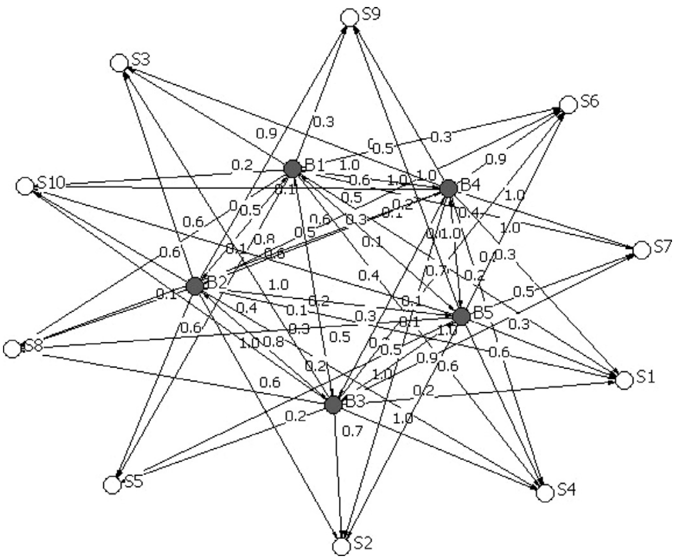

Fig. 7. Relationships among 5 buyers and 10 sellers after 50 rounds of sourcing.

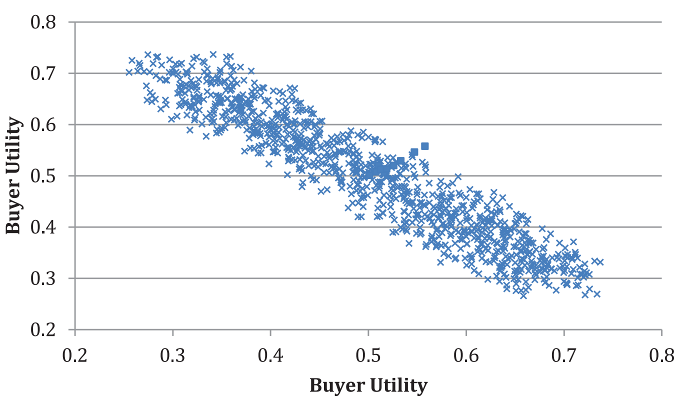

Fig. 8. Efficient points in negotiations.

zone. In addition, the negotiation results that are suggested by the boundary-spanning service are all located near the 45-degree horizon. These imply that the final agreement is both efficient and fair to buyers.

4.5. Further discussions about buyers' preference and sellers' discount quantity

In this study, preference indications are a crucial aspect of the boundary-spanning service, as they are the basis for the service engine's recommendations. We choose two criteria (price and quantity) to show how preferences could affect a company's decisions. We first define two kinds of buyers: price-prone (those prone to make decisions based on price) and deal-prone (those prone to make decisions based on quantity) (Granzin, 1981; Wilkes & Valencia, 1985). For price-prone buyers, the price preference value should be higher than 0.5 in their preference vectors. Similarly, for deal-prone buyers, the quantity preference value should be greater than 0.5. We randomly generate five buyers in a price-prone group (Group 1) and another five buyers in a deal-prone group (Group 2). The preference vectors of the two groups of buyers are shown in Table 8.

Simulations are run where the two groups of buyers make purchases in ten separate periods, and other parameters remain unchanged. We use the average utility $\bar{U}$ to represent the utilities of all buyers in a group, and the number of cooperative purchases N_{cp} to evaluate cooperation in the group. The results are shown in Table 9.

Table 8
Two groups of buyers with primary preference vectors.

Price proneness group (Group 1)	Deal proneness group (Group 2)
(0.545,0.015,0.352,0.079,0.010)	(0.089,0.194,0.058,0.602,0.058)
(0.551,0.138,0.129,0.038,0.144)	(0.133,0.022,0.125,0.550,0.170)
(0.650,0.013,0.034,0.167,0.137)	(0.157,0.164,0.049,0.500,0.170)
(0.532,0.085,0.145,0.119,0.119)	(0.052,0.122,0.047,0.684,0.095)
(0.540,0.076,0.221,0.155,0.008)	(0.227,0.010,0.154,0.501,0.109)

Table 9
$\bar{U}$ and N_{cp} for two different groups.

Average utility $\bar{U}$		Number of cooperative purchases N_{cp}	
Group 1	Group 2	Group 1	Group 2
0.51	0.59	2	0
0.67	0.70	2	2
0.64	0.56	2	0
0.60	0.71	0	2
0.58	0.52	2	0
0.64	0.61	3	0
0.60	0.67	2	2
0.56	0.59	2	0
0.53	0.57	0	0
0.58	0.56	2	2
Avg = 0.591	Avg = 0.608	Avg = 1.7	Avg = 0.8

From this, we can see that there is no significant difference in average utility between price-prone buyers (0.591) and deal-prone buyers (0.608). However, it is interesting to note that there is less cooperative purchasing in the deal-prone group (0.8) than in the price-prone group (1.7). This may be due to the high level of sensitivity among deal-prone buyers to their order quantity. An increase in order quantity might significantly reduce their utility, and therefore they have lower incentives to make cooperative purchases.

4.6. Further discussions about sellers' discount quantity

We assume that all sellers' discount quantities are distributed normally with mean $\bar{q}_0$ and variance σ^2, and that all buyers' order quantities for each purchase period are distributed normally with mean $\bar{q}$ and variance σ^2. To establish how both $\bar{q}_0$ and $\bar{q}$ affect the decisions of buyers, we examine three different market scenarios involving 10 buyers and 20 sellers: (1) $\bar{q} = \bar{q}_0/4 = 375$, $\sigma^2 = 100$ represents the scenario in which the sellers' discount quantity is too high in comparison to the buyers' order quantity; (2) $\bar{q} = \bar{q}_0/2 = 750$, $\sigma^2 = 100$ represents the medium scenario; and (3) $\bar{q} = \bar{q}_0 = 1500$, $\sigma^2 = 100$ represents the situation in which the sellers' discount quantity is very low in comparison to the buyers' order quantity. The experiment is repeated 100 times.

Fig. 9 shows that the number of cooperative purchases among the 10 buyers differs across the 3 scenarios. When the discount quantity of the sellers is much higher than the order quantity of the buyers (scenario 1), the buyers have fewer incentives to cooperate with other buyers by making collective purchases. When a seller's discount quantity is very high, it is hard for a buyer to find enough collaborators to reach the threshold required for the seller's discount. If they cannot find enough collaborators, they will give up on obtaining the discount, as too large an increase in their combined order quantity will also reduce the aggregate utility of buyers. Clearly, this situation has a more significant influence on deal-prone buyers than on price-prone buyers. As the sellers' discount quantity $\bar{q}_0$ falls to only twice the mean buyers' order quantity $\bar{q}$, the number of cooperative purchases increases rapidly. However, when $\bar{q}_0$ equals $\bar{q}$ in scenario 3, the number of cooperative purchases in each purchase period falls, and is even lower than in scenario 1. When sellers' discount quantities and buyers' order quantities are normally distributed with similar means,

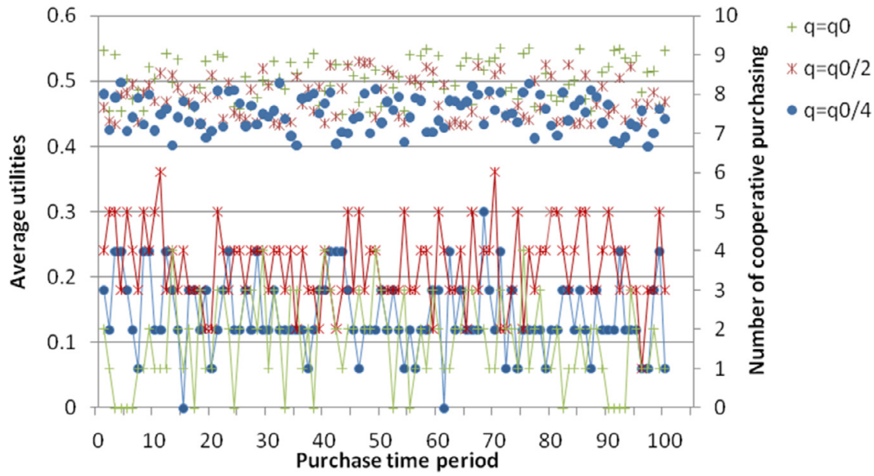

Fig. 9. Average utility and number of cooperative purchases in three different scenarios.

many of the order quantities reach the level of sellers' discount quantities, enabling the buyers to obtain a quantity discount directly from their seller without having to work with their competitors.

We can also observe from Fig. 9 that the average utility of the 10 buyers differs across the three scenarios. The average utility of buyers is lower in scenario 1 than in 2, as $\bar{q}_0$ is far higher than $\bar{q}$ in scenario 1, so fewer buyers cooperate with others and fewer can obtain quantity discounts from sellers. In contrast, buyers have higher average utilities in scenario 3 than in 2, as more buyers can obtain a quantity discount without having to cooperate with others, although the number of cooperative purchases is lower in scenario 3.

5. Conclusion

In this study we propose a boundary-spanning service for collaborative networks. Efficient access to the information stored in the cloud enables companies to adopt strategies to compete and/or cooperate with others. The boundary-spanning service can then help companies negotiate with other participants. Specifically, we have designed an agent that holds company decision makers ultimately responsible for the quality of their decisions. The boundary-spanning service is an intelligent intermediary that helps companies search for, inquire about, and recommend potential collaborators (sellers and/or buyers), based on preferences defined according to criteria such as price and delivery date. The agent can also negotiate with the company's competitors to gauge whether or not the utilities of both parties can be enhanced through cooperation. We demonstrate that companies can find qualified buyers or sellers using the boundary-spanning service. Through cooperative purchasing, companies are able to enhance their utilities by evaluating different strategies. Moreover, a decision maker could manage the evolution of the relationships via the collaboration established.

6. Discussion

The growth of the Internet has provided a platform for a new wave of supply chain collaboration. The method of defining collaborators as upstream, downstream, suppliers, and buyers has changed due to the dynamic nature of the platform. The decision to compete or collaborate depends on a company's strategies and business logics. Thus, intelligent systems play more important roles in managing "coopetition"—a revolutionary mindset combining cooperation and competition that allows companies to work with potential collaborators based on their preferences.

The environment expects an intelligent system that can dynamically manage profiles and collaborator relationships. The boundary-spanning agents are intelligent intermediaries that help to locate information, provide for contingencies, process discretionary services, and interact with collaborators intensively under the instruction of built-in business logics and strategies. In this study, we propose a framework using the boundary-spanning service agent for corporate coopetition. We use active documents to manage the business logics and strategies for negotiation where multiple-criteria decision-making problems are normally involved.

Future studies can adopt this framework to build intelligent systems that support complex business logics and sophisticated negotiation processes. As demonstrated, using active documentation can be a good option. In this study, we do not demonstrate how to reach the Pareto frontier, which demonstrates an efficient point. Future research can improve on the negotiation process.

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